

Correctness Detection and Answer Control Are Dissociable in Language Model Hidden States

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Abstract

Language models produce confident wrong answers whose output statistics are indistinguishable from confident correct answers by construction: when both have low output entropy, no output-space signal—entropy, calibration, self-consistency—can separate them. We show that *hidden states carry this information*, and that the hidden-state direction which best separates correct from wrong confident answers is **not** the direction that causally controls which answer the model generates.

We introduce the **bilateral oracle protocol**, a two-pass behavioral labeling design that isolates the confabulation-detection task by construction, without accessing hidden states. In the entropy-matched zone where output statistics are blind, a Fisher+PCA64 linear probe achieves AUROC = 0.854 and cross-validated AUROC = 0.7629 ± 0.0120 ($N = 500/\text{class}$; confirmed). CO labeling extends detection to all four tested architectures (Gemma = 0.8368, Mistral = 0.8580).

To test whether this detection geometry *controls* answer selection, we patch along the Fisher LDA class-mean axis across all layers and patch strengths. Max $\Delta F1 = +0.0004$ —indistinguishable from zero. Consistent with Cox et al. [2026]: the correctness-reading axis and the answer-writing axis are empirically dissociable—one reads whether an answer is correct; the other writes which answer gets generated.

1 Introduction

A language model that answers confidently and wrongly is more dangerous than one that answers uncertainly—not because the error is larger, but because it is invisible. When output entropy is low, no output-space signal can separate correct from wrong answers: they are entropy-matched by construction. The safety question then moves inside: *do hidden states carry the information that output distributions discard?*

This paper pursues a precise version of that question and an equally precise follow-up: if hidden states can detect confabulation, does the detection geometry also causally control which answer is generated? We find the answer to the first question is yes, and the answer to the second is no. These two facts together are more informative than either alone: the correctness-reading axis and the answer-writing axis are **empirically dissociable**.

Contributions.

1. **Bilateral oracle protocol.** A two-pass labeling design that assigns CC (confident-correct) and CW (confident-wrong) labels in the entropy-matched zone where output statistics are blind by construction. The labeling is behavioral and estimator-agnostic.
2. **Detection result.** Fisher+PCA64 at L26 step-1 achieves AUROC = 0.854 for CC vs. CW on Qwen2.5-1.5B-Instruct; cross-validated at AUROC = 0.7629 ± 0.0120 ($N = 500/\text{class}$, 5-fold). CO labeling extends this to four architectures.
3. **Causal dissociation.** Class-mean centroid patching along the Fisher LDA direction (EXP-H) produces $\max \Delta F1 = +0.0004$ across $\lambda = 0.1\text{--}2.0$ and all probed layers—the detection direction is not the causal control direction.

2 Method

2.1 Bilateral Oracle Protocol

The bilateral oracle assigns knowledge-source labels through two independent inference passes. **CC (confident-correct)** items: the model generates a correct answer ($F1 \geq 0.50$) and the output entropy is below a model-calibrated threshold (θ_{conf}). **CW (confident-wrong)** items: the model generates an incorrect answer with the same entropy constraint.

The entropy threshold θ_{conf} is calibrated per-model to yield a balanced pool. For Qwen2.5-1.5B-Instruct on TriviaQA, $\theta_{\text{conf}} = 1.1043$ nats. Hidden states are always collected from the same inference pass. Items that do not meet either criterion are excluded.

Why entropy-matching matters. If CC items happened to have lower entropy than CW items, a trivial entropy threshold would solve the task. Entropy-matching rules this out by construction: after matching, the residual entropy difference is only 0.037 nats—the entropy baseline achieves AUROC = 0.614 on this matched set. Any Fisher signal above this threshold is genuinely hidden-state-specific.

2.2 Fisher+PCA64 Probe

All probes follow the pipeline:

$$\text{PCA}(n=64) \rightarrow \text{LDA}(\text{solver} = \text{lsqr}, \text{shrinkage} = \text{auto})$$

applied to hidden states at layer $L = 26$ (or the penultimate layer for non-28-layer models), generation step $s = 1$ (immediately before the first output token is decoded). Layer and step are selected from prior sweeps; they are hyperparameters of the measurement, not universal properties. PCA to 64 components addresses Fisher LDA’s degenerate covariance pathology at high dimension / low sample ratio ($d = 1536\text{--}3072$, $N \approx 100\text{--}500/\text{class}$).

2.3 CO Labeling (Universal L2 Framing)

For architectures with very low output entropy, the narrow entropy-matched window contains too few items for reliable measurement. CO (Correctness Oracle) labeling uses a simpler framing: all items with output entropy $\leq \theta_{\text{conf}}$ are labeled CC (correct) or CW (wrong), without requiring tight entropy balance between classes. This recovers L2 detection for all tested architectures. The Fisher signal gap is statistically equivalent under both framings (CO gap = 0.109, entropy-matched gap = 0.124, $p = \text{n.s.}$).

3 Results

3.1 Confabulation Detection at L2

Fisher+PCA64 achieves AUROC = 0.854 on the entropy-matched CC vs. CW task (Qwen2.5-1.5B-Instruct, TriviaQA, $N = 100/\text{class}$, shuffled control = 0.426). The gap over the entropy baseline is +0.240 AUROC (Fisher 0.854 vs. entropy 0.614). Large- N cross-validation ($N = 500/\text{class}$, 5-fold) yields $\text{CV} = 0.7629 \pm 0.0120$, range 0.744–0.781. All folds have clean shuffled controls.

Llama-3.2-3B-Instruct replication: Fisher = 0.818, entropy = 0.453, gap = 0.365. The gap is *larger* on Llama because the entropy window is more tightly matched.

Table 1: L2 confabulation detection: Fisher+PCA64 vs. output-space baselines (Qwen2.5-1.5B-Instruct, TriviaQA, $N = 100/\text{class}$, entropy-matched). **TODO:** B1–B3 values pending `baseline_comparison_v1` run.

Method	AUROC	Shuffled	Notes
Fisher+PCA64 (ours)	0.854	0.426	Hidden state, L26 step-1
B1: Verbalized uncertainty	<i>TBD</i>	—	Output-space only
B2: Self-consistency ($N=5$)	<i>TBD</i>	—	Output-space only
B3: Top-1 probability	<i>TBD</i>	—	Output-space only
Entropy baseline	0.614	≈ 0.50	Matched by design

Table 1: Fisher vs. output-space baselines at L2.

3.2 Architecture Generalization (L1)

To establish the bilateral oracle protocol across architectures, we first confirm Fisher+PCA64 signal at L1 (knowledge-source routing: parametric vs. context-dependent items) across five independent architectures. All five architectures exceed the pre-registered floor of $\text{AUROC} \geq 0.70$ with clean shuffled controls (Table 2). Architecture spread is 0.731–0.846; Phi-3.5-Mini (0.846) is highest.

Table 2: L1 bilateral oracle Fisher+PCA64 AUROC across five architectures (TriviaQA, $N = 197\text{--}200/\text{class}$).

Model	Family	AUROC	Shuffled	CI 95%
Qwen2.5-1.5B-Instruct	Qwen	0.731	0.598	[0.63, 0.83]
Llama-3.2-3B-Instruct	Llama	0.746	0.502	[0.65, 0.83]
Gemma-2-2B-IT	Gemma	0.753	0.530	[0.65, 0.85]
Mistral-7B-Instruct	Mistral	0.778	0.552	[0.69, 0.86]
Phi-3.5-Mini-Instruct	Phi	0.846	0.499	[0.76, 0.92]

Scope note. At L1, output entropy achieves 0.87–0.90 AUROC (vs. Fisher 0.73–0.85): Fisher is not uniquely essential at L1. The signal at L2 is where Fisher becomes irreplaceable: it adds 0.240–0.365 AUROC over entropy in the entropy-matched confident zone.

3.3 CO Labeling Extends L2 to Four Architectures

For Gemma-2-2B-IT and Mistral-7B-Instruct-v0.3, the entropy-matched framing degenerates (fewer than 10 items per class in the narrow entropy window). CO labeling recovers detection for all four tested architectures:

Table 3: CO-labeled L2 confabulation detection across four architectures. Gap = Fisher AUROC – Entropy AUROC.

Model	Fisher	Entropy	Gap	Verdict
Qwen2.5-1.5B-Instruct	0.885	0.653	+0.232	CO_RECOVERS
Gemma-2-2B-IT	0.837	0.631	+0.206	CO_RECOVERS
Mistral-7B-Instruct-v0.3	0.858	0.647	+0.211	CO_RECOVERS
Llama (CV, $N = 500$)	0.763	—	+0.163	CV STABLE

The CO gap (0.163–0.232) is consistent with the entropy-matched gap (0.240/0.365), confirming that entropy matching is not required to observe the L2 signal—it is a genuine geometric property of the residual stream, not a framing artifact.

4 Causal Dissociation

4.1 EXP-H: Centroid Patching Null

The detection result (AUROC = 0.854) establishes that the Fisher LDA class-mean direction $\mathbf{v}_{\text{Fisher}}$ separates CC from CW items in representation space. EXP-H tests whether *patching* along $\mathbf{v}_{\text{Fisher}}$ at inference time can convert CW items to CC-like answers.

Protocol. For each probed layer ℓ and patch strength $\lambda \in \{0.1, 0.5, 1.0, 1.5, 2.0\}$, we add $\lambda \cdot (\mu_{\text{CC}} - \mu_{\text{CW}})$ to the residual stream at step $s = 1$, where μ_{CC} and μ_{CW} are the layer- ℓ class-mean vectors estimated from the training split. We then complete generation and measure the change in F1: $\Delta\text{F1} = \text{F1}_{\text{patched}} - \text{F1}_{\text{original}}$.

Result. Max $\Delta\text{F1} = +0.0004$ across all probed layers and all λ values—indistinguishable from zero. The detection direction is **epiphenomenal**: it reliably identifies which items are CC vs. CW, but it is not the mechanism by which the model selects its answer.

4.2 Two-Axis Interpretation

Cox et al. [2026] demonstrated that the *committed-answer direction*—the residual-stream direction encoding the model’s specific output label—achieves ~ 0.90 AUC before any chain-of-thought token is generated, and that steering along this direction flips answers in $>50\%$ of cases. These results are consistent with ours rather than contradictory.

The geometry has (at minimum) two distinct axes:

- **The detection axis** (Fisher LDA class-mean direction): encodes whether an answer is correct. Readable with AUROC = 0.854. Not writable via class-mean patching.
- **The control axis** (committed-answer direction, Cox et al.): encodes which specific answer is generated. Writable: steering flips answers in $>50\%$ of cases.

EXP-H establishes that the detection axis does not serve as the control axis. This is a sharper claim than “epiphenomenal”: the signal is real and detectable; the causal mechanism of answer selection lies on a different geometric axis.

Deployment implication. Gate-3 confabulation detection should be treated as a *routing signal*: when the Fisher score indicates CW, route to a fallback action (refuse, retrieve, or escalate). The null patching result does not diminish the detector’s value for routing; it clarifies that correction requires targeting the control axis, not the detection axis.

5 Related Work

Uncertainty estimation. Semantic entropy [Kuhn et al., 2023] generates multiple samples to estimate uncertainty from output-level consistency. Self-knowledge elicitation [Kadavath et al., 2022] prompts for verbal confidence. Semantic entropy probes [Kossen et al., 2024] extend semantic entropy to hidden-state probes. The key distinction from the present work: all three operate on the full uncertainty spectrum, not within the confident zone where output statistics are blind by construction. The bilateral oracle specifically targets the entropy-matched subpopulation where these baselines fail.

Knowledge-source probing. Tighidet et al. [2024] train hidden-state classifiers to distinguish parametric from contextual knowledge using contradiction prompts. The bilateral oracle differs: labels are assigned by context-withholding (behavioral intervention), not prompt injection—avoiding the confound that the probe may detect injection style rather than epistemic state.

Causal probing. Inference-Time Intervention [Li et al., 2023] achieves directional control of truthfulness via steering. Cox et al. [2026] show the committed-answer direction achieves high AUC and causal control before chain-of-thought generation. Yuan et al. [2026] find that probes achieving 0.95 AUROC for CoT reasoning errors fail to correct them under four intervention methods—directly paralleling our EXP-H null result in the confabulation domain. The pattern is emerging as a general principle: detection geometry $\neq$ control geometry.

Probing methodology. Probing classifiers [Alain & Bengio, 2016] and the linear representation hypothesis [Elhage et al., 2022] establish the framework. The bilateral oracle contribution is the two-pass labeling design, not the probe architecture: it isolates knowledge-source from correctness without requiring internal access.

6 Limitations

Difficulty confound (Q14, unresolved). The bilateral oracle excludes items with ambiguous F1 scores (nocontext $F1 \in (0.05, 0.50)$). Items passing the PARAM filter may be systematically easier, and Fisher may partly detect item difficulty rather than epistemic state. A within-PARAM probe separating easy vs. hard items achieves $AUROC = 0.640$ ($CI = [0.43, 0.66]$, $N = 73$), but the CI overlaps the shuffled range. The difficulty confound is not resolved; Q14v4 (TriviaQA train split, tier-targeted) will address it with larger N per tier.

Dataset scope. All results are on TriviaQA. The probe transfers to HotpotQA ($AUROC = 0.657$, 84.5% relative efficiency) but not to MMLU-STEM ($\approx$ random). All observability claims are **format-scoped**: the bilateral oracle learns a geometry specific to the TriviaQA question-answer format.

Goldilocks zone. The bilateral oracle requires instruction-following capability to generate both CC and CW items with low entropy. Pure base language models and models too small for the task (Qwen2.5-0.5B: zero qualifying PARAM items) fall outside the zone. All results are restricted to the 1B–8B instruction-tuned range tested.

Single control experiment. EXP-H uses class-mean centroid patching. Attention-head patching and SAE feature patching along the Fisher direction remain untested. The null result is specific to the centroid-direction intervention; more targeted patching may produce non-null effects.

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